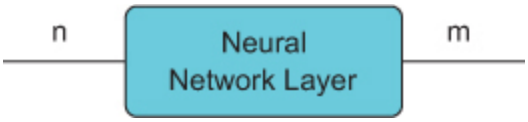
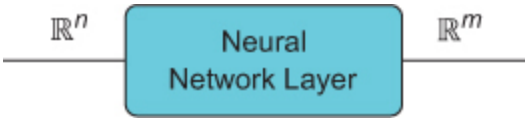
complex diagrams, as long as all the strings' types are compatible. For example, here's a single layer of a neural network as a string diagram:



Reading from left to right, we see that some data of type `n` flows into a process box called Neural Network Layer and produces output of type `m`. Since neural networks typically take vectors as inputs and produce vectors as outputs, these types refer to the dimensions of the input and output vectors respectively. That is, this neural network layer accepts a vector of length or dimension *n* and produces a vector of dimension *m*. It's possible that $n = m$ for some neural network layers.

This manner of *typing* the strings is simplified, and we do it only when it's clear what the types mean from the context. In other cases, we may employ mathematical notation such as $\mathbb{R}$ for the set of all real numbers, which in programming languages basically translates to floating-point numbers. So for a vector of floating-point numbers with dimension *n*, we could type the strings like this:



Now that the typing is richer, we not only know the dimensions of the input and output vectors, we know that they're real/floating-point numbers. While this is almost always the case, sometimes we may be dealing with integers or binary numbers. In any case, our Neural Network Layer process box is left as a black box; we don't know exactly what's going on in there other than the fact that it transforms a vector into another vector of possibly different dimensions. We can decide to zoom in on this process to see what specifically is happening: